

Finals Task 2. Navigating Power BI

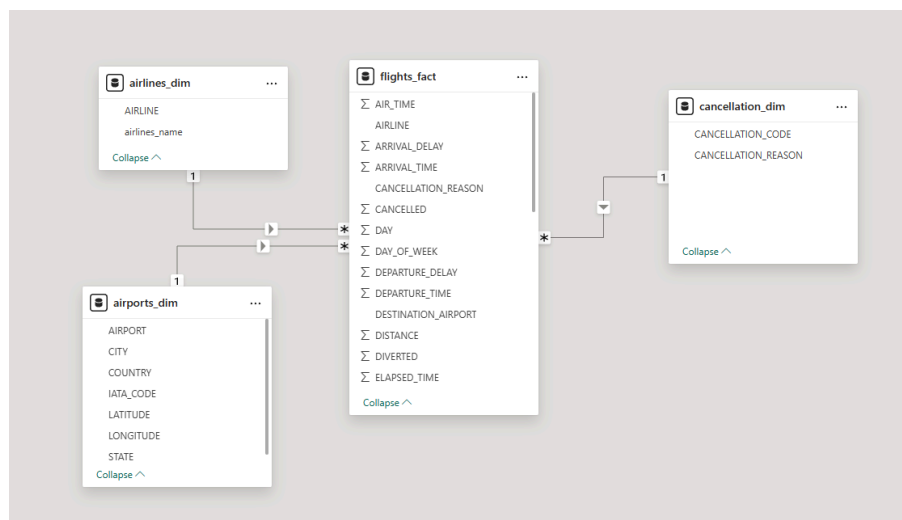
Description:

This task focused on learning the complete Power BI workflow, including data preparation, data modeling, analysis, and visualization. The activity used the flights_data and airports datasets to create a structured and interactive business intelligence report.

The datasets were first cleaned and prepared by fixing column labels, handling blank values, removing unnecessary fields, and organizing the data into one fact table and multiple dimension tables. Additional tables such as airline, cancellation reasons, and airports were separated to create a proper star schema data model.

After building the data model, DAX calculated columns and measures were created to analyze flight performance. Metrics such as total flights, canceled flights, delayed flights, and on-time flights were calculated, along with their corresponding percentages. Measures were also organized into folders to maintain a cleaner and more structured model view.

Different Power BI visuals were then created to present the computed insights through dashboards and reports. Formatting tools and visualization features were explored to improve the overall presentation and readability of the final dashboard. This activity strengthened skills in Power BI, data modeling, DAX calculations, and dashboard design.



US AIRLINES FLIGHT INFORMATION AS OF 2015

1M

Total Flights



443K

Delayed Flights

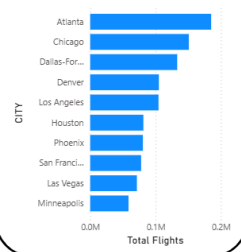


20K

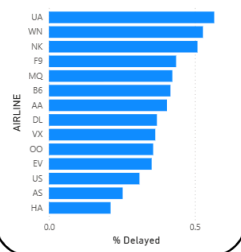
Canceled Flights



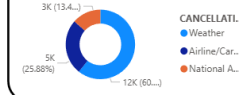
Total Flights by CITY



% Delayed by AIRLINE



Canceled Flights by CANCELLATION_REASON



% Canceled by DAY_OF_WEEK



% On-Time

5...

% Delayed

42...

% Canceled

2%

Total Flights by STATUS

STATUS: Cancelled, Delayed, On time

